

[14] Nomic Embed: Training a Reproducible Long-Context Text Embedder

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1-Minute Executive Takeaway

Key Finding: Validates nomic-embed-text as a high-accuracy, 8k-context open-weight embedding model for document retrieval.

Core Research Objective

To train and release a high-performance, fully reproducible, open-weight text embedding model with an extended context window (8,192 tokens).

Methodology & Architecture

Multi-stage contrastive pre-training and fine-tuning on diverse text retrieval pairs, producing 768-dimensional dense vector embeddings optimized for sentence and paragraph similarity.

Key Empirical Findings & Evidence

Achieved top-tier retrieval performance on the MTEB benchmark, outperforming OpenAI `text-embedding-ada-002` while supporting long-context document chunks.

Direct Relevance to Our Kwiz Paper

Cited in Section 3.1 and Section 4.8. `nomic-embed-text` is the dedicated embedding backbone of Kwiz, enabling semantic retrieval over curriculum slide chunks with sub-millisecond similarity computation.